



Institute of Computer Engineering Technology



iCET Certified Developer

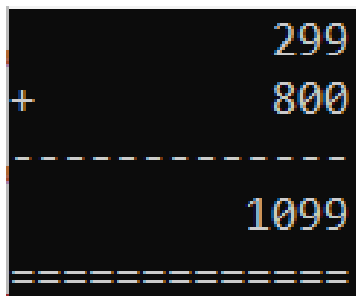
ASSIGNMENT

Module	Programming Fundamentals
Assignment Title	Variable and Data Types
Assignment Type	Written Assignment

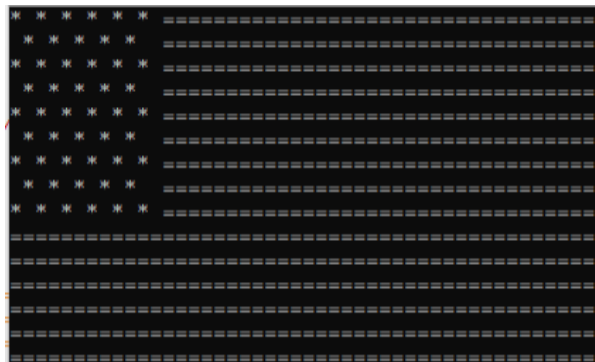
01. Write a java program to print your name , address, phone number and email.
02. Write a java program to obtain the following result.



03. Write a Java program using only "System.out.println()" to get following output.



04. Write a Java program to print an American flag on the screen.



05. What are the output of following program?

Program A

```
class Example{
    public static void main(String args[]){
        System.out.println("Institute of Computer Engineering Technology");
        System.out.println("223 A,");
        System.out.println("Galle Road,");
        System.out.println("Panadura.");
    }
}
```

Program B

```
class Example{
    public static void main(String args[]){
        System.out.println("Institute of Computer Engineering Technology");
        System.out.print("223 A,");
        System.out.print("Galle Road,");
        System.out.print("Panadura.");
    }
}
```

06. What is the difference between print() and println()? Explain your answer with suitable examples.

07. What is the different between letter a of line 1 and line 2?

```
class Example {
    public static void main(String args[]){
        System.out.println("a");    //line 1
        System.out.println('a');    //line 2
    }
}
```

08. What is the different between 7 of lines 1 , 2 , 3 and 4 ?

```
class Example {
    public static void main(String args[]){
        System.out.println(7);    //line 1
        System.out.println(7.0);  //line 2
        System.out.println("7");  //line 3
        System.out.println('7');  //line 4
    }
}
```

09. What are the type of java literals? Explain with a suitable example.

10. What will be the output when you compile and run the following program? Explain your answer.

```
class Example {
    public static void main(String args[]){
        System.out.println(1100100);    //line 1
        System.out.println(0B1100100);  //line 2
        System.out.println(01100100);   //line 3
        System.out.println(0x1100100);   //line 4
    }
}
```

11. Which of the following lines are valid statements? Explain your answers.

```
class Example{
    public static void main(String args[]){
        System.out.println(0B11100100); //Line 1
        System.out.println(0b11100100); //Line 2
        System.out.println(0B11100200); //Line 3
        System.out.println(0144); //Line 4
        System.out.println(0148); //Line 5
        System.out.println(0x64); //Line 6
        System.out.println(0xabc); //Line 7
        System.out.println(0xabcg); //Line 8
        System.out.println(0X); //Line 9
        System.out.println(0Xffffff); //Line 10
    }
}
```

12. What is the result of attempting to compile and run the program?

```
class Example{
    public static void main(String args[]){
        System.out.print("A\nB");
        System.out.println("CD");
        System.out.print("\nEF\n");
        System.out.print("G\nH")
    }
}
```

13. Write a java program using single "System.out.print()" to get following output.

i. iCM – iCET CERTIFIED MASTER

ii. iCM
 iCET
 CERTIFIED
 MASTER

iii. iCM
 iCET CERTIFIED MASTER

14. What is the output of following program.

```
class Example {
    public static void main(String args[]){
        System.out.println("Hello\nJAVA");
        System.out.println("Hello\tJAVA");
        System.out.println("Hello\bJAVA");
        System.out.println("\ \Hello JAVA\ ");
        System.out.println("\"Hello\nJAVA\"");
        System.out.println("'Hello\nJAVA'");
    }
}
```

What are the Escape Sequences used in above program? Explain about them with suitable example.

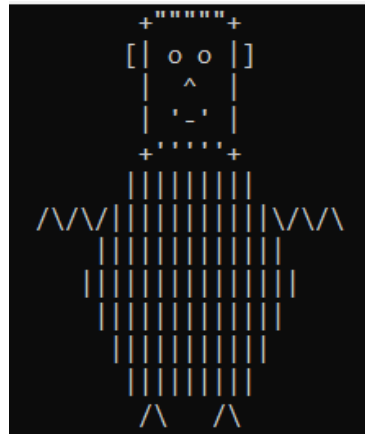
15. Write java statements to get the following output. (Only one sout statement can be used).

- a. Java is a typed language
- b. AB"CB
- c. AB\CD
- d. C:\Windows\Program
- e. AB\"CD
- f. AB\\\"CD
- g. AB\nCD
- h. AB\tCD
- i. AB\bCD

16. Write a java program to get following output.

```
( ) / _ _ _ | _ _ _ | _ _ _ |
| | | | | | | | | |
| | | | | | | | | |
| | \ _ _ _ | _ _ _ | | |
```

17. Write a program in Java to display following picture.



18. Explain line 1 , 2 and 3 in following program.

```
class Example{
    public static void main(String[] args){
        int i;                //line 1
        i=103;                 //line 2
        System.out.println(i); //line 3
    }
}
```

19. Write a java program create 2 variables as integer, named x , y.

- i. assign values x – 102 y – 103
- ii. Print the values as x , y format
102 103
- iii. Print the values in y , x format.
103 102

20. What is the result of attempting to compile and run the program?

```
class Example{
    public static void main(String args[]){
        int x;
        System.out.println(x);
    }
}
```

- i. prints : nothing ii. prints : x iii. prints : 100
iv. Runtime error v. Compile time error

21. What is the result of attempting to compile and run the program?

```
class Example{
    public static void main(String args[]){
        int x=10,y=20,z=30;
        System.out.println(x);
        System.out.println(y);
        System.out.println(z);
        int y=200;
        System.out.println(y);
    }
}
```

22. What will be the output when you compile and run the following program? Explain your answer.

```
class Example {
    public static void main(String args[]){
        int x,y;
        x=100;
        y=200;
        System.out.println(x);
        System.out.println(y);
        y=300;
        System.out.println(x);
        System.out.println(y);
    }
}
```

23. Which of the following can be legally inserted at line 12? Explain your answer.

```
class Example{
    public static void main(String args[]){
        int x;
        //Line 12
        System.out.println(x);
    }
}
```

- A. x+1; B. x=100; C. int y=100 D. int x=200; E. Insert nothing.

24. What is the result of attempting to compile and run the program?

```
class Example{
    public static void main( String args[] ){
        int x,y,z; //line 1
        x=y=z=10; //line 2
        int a=12,b,c=4; //line 3
        int p=10; //line 4
        int q=p; //line 5
        int i=j=k=10; //line 6
    }
}
```

- A. Compile error at line 2 B. Compile error at line 3
C. Compile error at line 5 D. Compile error at line 6
E. None of the above

25. What are the output of following program and Explain your answer.

```
class Example{
    public static void main(String args[]){
        System.out.println("10"+"20");    //line 1
        System.out.println(10+20);        //line 2
    }
}
```


26. What will be the output when you compile run the following program?

```
class Example{
    pubic static void main(String args[]){
        System.out.println(10+20+30);
        System.out.println("10+20+30");
        System.out.println("10+20"+30);
        System.out.println("10"+"20"+"30");
        System.out.println("10"+20+30);
        System.out.println(10+20+"30");
        System.out.println(10+"20"+30);
    }
}
```

27. What is the output for following code fragment and explain your answer?

- A. System.out.println(1+2+3);
- B. System.out.println("1"+"2"+"3");
- C. System.out.println('1'+2+'3');
- D. System.out.println('1'+ " "+2+' " '+3');
- E. System.out.println('A'+B+'C');
- F. System.out.println("A"+"B"+"C");
- G. System.out.println('A'+100+200);
- H. System.out.println('A'+ " "+B+' " '+C');

28. What is the different between Question 20 program and following program.

```
import java.util.*;
class Example{
    public static void main(String[] args){
        int i;
        i=input.nextInt();
        System.out.println(i);
    }
}
```

29. Write a java program using the help of the "Scanner" ,
- Input two integer values and store them in two variables.
 - Print the values as they assign to the variable and finally print them as following.
1st input -12
2nd input -34
 - Declare an integer variable and initialize by sum of the above user input and print following.
 $12 + 34 = 46$

30. Write a Java program that takes an integer (n) and display the output as below to differentiate the arithmetic addition and string concatenation.
- Sample Output:-
- Input number: 8
888
24

31. Write a java program to get four subjects marks and print output as below.
- | | |
|----------------|-------|
| Combined Maths | - 81 |
| Chemistry | - 72 |
| Physics | - 84 |
| English | - 78 |
| Total | - 315 |

32. Write a Java program to invert the RGB values input by the keyboard. Colour inversion is the colour left over when you deduct one RGB value from 255.
- Sample Output:-
- Enter Red value : 165
Enter Green value : 170
Enter Blue value :221
Inversion of given colour - [90, 85, 34]

33. Fill the blank following table about primitive data type.

Data Type	Size	Description
byte	1 byte	
short		-32 768 to 32 767
int		
long		
float		
double	8 bytes	
boolean		
char		

34. Which of the following identifiers are not valid?

myVariable	@name	Customer	getFirstName
new	user_input	main	longVariable
MyClass	user-input	\$percent	_piece
My Variable	Boolean	123num	java.org

35. Study following code and explain what are the different between line 1 and line 2.

```
class Example{
    public static void main(String[] args) {
        System.out.println("iCET"); //line 1
        String s="connect intelligents...";
        System.out.println(s); //line 2
    }
}
```

36. Write a program that asks the user's name and then greets the user by name. For example, if the user's name is Amal, then the program should display "Hello Amal, nice to meet you!"

```

37.    class Example {
        public static void main(String[] args) {
            int x = 10;
            int y = 5;
            String z = x + 'y' + "y" + x + y + 'z';
            System.out.println("z is: " + z);
        }
    }

```

What is the output of this program?

- A) z is: 10y5y105z B) z is: 105yz
 C) z is: 205yz D) z is: 15yy105z

38. Write a java program to get you'r best friend details as input and print small description like below. Numbers should get as int value.

```

Enter your friend's name: Kasun
Enter your friend's age: 21
Enter your friend's living place: Panadura

Kasun is my best friend. He is 21 years old and lives in the beautiful town of Panadura.

```

39. Write a java program to get A/L student subject name and subject marks and print total as following.

```

----Subject total calculator----

Subject 1 name - Maths
Maths marks - 67
Subject 2 name - Chemistry
Chemistry marks - 59
Subject 3 name - Physics
Physics marks - 81

Maths          67
Chemistry      59
Physics        81

Total          207

```

40. Complete the following program?

```
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input=new Scanner(System.in);
        System.out.print("Input number 1 : ");
        int num1=input.nextInt();
        System.out.print("Input number 2 : ");
        int num2=input.nextInt();
        System.out.println(num1+" "+num2);
        //-----Do not modify before this line-----
        //Insert java code(s) here to swap the two
        //variables "num1" and "num2"
        //as an example when you input 10 for
        //num1 and 20 for num2, then
        //the first output should be "10 20" and the
        //second output "20 10"
        //-----Do not modify after this line-----
        System.out.println(num1+" "+num2);
    }
}
```